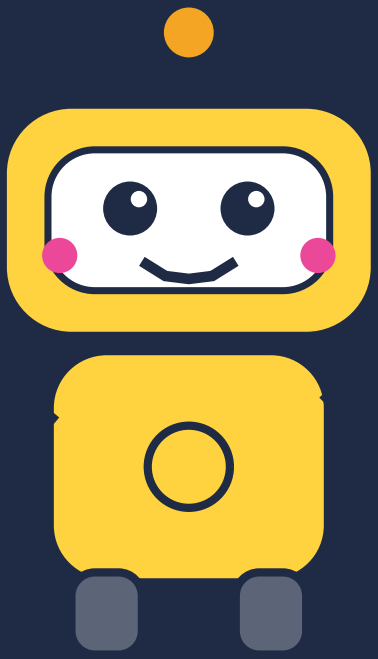
TEACHER BOOK

Think Like a Coder

Teacher book for Class 8

Period-by-period plans for every CT and AI chapter
Teacher guides for both 20-hour projects
Answer keys, assessment guide and photocopiable journal



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 8 · Follows NCERT Class 8 maths, CBSE's four AI chapters and two projects

SAMPLE EDITION · DRAFT

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About this teacher book

This book goes with the *Think Like a Coder* Class 8 workbook. It gives you a plan for every chapter, so that a maths or class teacher can teach computational thinking and AI confidently without being a computer specialist. CBSE suggests about **100 hours** a year for Class 8; the plans here use **150 periods of 40 minutes**, which is about 100 hours.

★ What is in this book

- ✓ **A year plan** showing which chapter to teach in which term, next to the NCERT maths chapter.
- ✓ **A period-by-period plan for every chapter**, including the AI chapters: what you do and what students do.
- ✓ **A hands-on activity** for every chapter, with set-up steps and a tip.
- ✓ **Common mistakes** to watch for, and how to support or stretch students.
- ✓ **An exit check** with answers, and what to note in the observation journal.
- ✓ **The answer key** for every page of the student workbook.
- ✓ **Photocopiable** observation journal and term summary pages at the back.
- ✓ **A teacher guide for each 20-hour project**: week-by-week plan, roles, set-up, pitfalls and how to mark with the rubric.

✓ Before you start

- ✓ Read the chapter in the student book and these notes. Most lessons need 5 to 10 minutes to prepare.
- ✓ Get the kit items ready (listed under *At a glance*). Everything is reusable.
- ✓ Plan pairs in advance: mix confident readers with children who prefer talking and acting out.
- ✓ Teach the CT chapter in the same week or fortnight as the matching NCERT maths chapter, so the maths is fresh.
- ✓ Keep it active: children should talk, act out and build more than they listen.

Computers are optional

No computer lab is needed. Every CT and AI idea is taught with paper, cards and talk, and the no-code labs in CBSE's handbook have card versions here. Where computers are available, the optional boxes suggest running the labs with a no-code tool. A projector helps but is optional.

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How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.
2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.
3. **Modify** 8 min
   Children change the steps to do something new.
4. **Put in order** 8 min
   Jumbled code cards go back in the right order.
5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

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Year plan

7 CT chapters of 8 periods plus 4 review periods (about 40 hours), 4 AI chapters of 7 periods plus 2 review periods (about 20 hours), and two projects of 30 periods each (about 40 hours): **150 periods** of 40 minutes in all. Teach each chapter alongside the NCERT maths chapter with the same name; the order and timing can follow your school's maths plan.

Ch	Chapter	Coder skill	Level	Periods	Term
1	A Square and a Cube	Test squares and cubes with prime factors and loops	C6	8	Term 1
2	Power Play	Exponents, doubling and counting combinations	C6	8	Term 1
3	A Story of Numbers	Number bases: binary, ternary and changing base	C6	8	Term 1
4	Quadrilaterals	Classify shapes with if-else rules	C6	8	Term 1
5	Number Play	Divisibility rules and remainders as quick tests	C7	8	Term 2
6	We Distribute Yet Things Multiply	The distributive property and identities for fast calculation	C7	8	Term 2
7	Proportional Reasoning	Ratios, the unitary method and scaling	C7	8	Term 2
AI1	AI Project Lifecycle	Define, collect, train, test and deploy	C8	7	Term 1
AI2	AI and Its Applications	AI for people and planet; training no-code models	C8	7	Term 1
AI3	Data and Fairness in AI	Spot unbalanced data and rebalance it	C8	7	Term 2
AI4	Ethics and Responsible AI	Privacy, checking facts and human accountability	C8	7	Term 2
P1	Sports for Everyone	Survey, analyse with ratios and recommend changes	C8	30	Term 1
P2	Fair AI Audit	Test a classifier for bias, rebalance and report	C8	30	Term 2
	Revision and term assessment	Review games, the term summary and parent feedback	All	6	Both

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Assessment and the observation journal

CBSE asks for continuous, competency-based assessment in CT&AI: written puzzles, group activities and a **Teacher Observation Journal**. This book makes that simple. In each chapter, the exit check and the 'what to note' box tell you what to look for; record one level per child per skill every few chapters, not every lesson.

Skill	Beginning	Developing	Secure	Extending
Algorithms	Follows given steps	Traces loops and conditions	Writes converters and classifiers	Designs, tests and improves algorithms
Number structure	Recalls facts	Uses rules with help	Uses bases, powers and identities	Explains why rules work
AI projects	Names the stages	Cleans data and measures accuracy	Runs the lifecycle and tests per group	Finds bias and fixes it with data
Projects	Takes part with help	Completes stages with help	Plans, tests and presents with the team	Improves the work using evidence

✓ How to assess without extra tests

- ✓ **Watch and listen** during activities: explaining, predicting and fixing show understanding better than a written answer.
- ✓ **Use the exit check** at the end of each chapter: two quick questions, answers in this book.
- ✓ **Compare with self-checks**: each chapter ends with 'How did I do?', where children colour a face for each 'I can' statement.
- ✓ **Summarise each term** on the term summary page: one level per area and a short comment for the report card.

The four levels

Use the four levels for every skill. **Beginning**: needs help to start. **Developing**: succeeds with prompts or a partner. **Secure**: succeeds alone on familiar problems. **Extending**: solves new problems and explains why. Most children will be Developing or Secure; that is expected and healthy.

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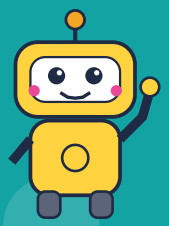
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TEACHER NOTES · CHAPTER 1 · LEVEL C6

A Square and a Cube

Coder skill: Test squares and cubes with prime factors and loops

Goes with NCERT maths, Class 8, chapter 1



At a glance	
Students will	test squares and cubes with prime factors, trace a checking algorithm and find patterns in squares and cubes.
Key words	**perfect square** (a number times itself), **perfect cube** (a number used three times), **prime factorisation** (a number as a product of primes), **square root** (the number that was squared)
Materials	Unit cubes or sugar cubes, prime factor cards, squared paper, trace tables
Time	8 periods of 40 minutes
Link	Teach alongside chapter 1 of the NCERT Class 8 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *perfect square*, *perfect cube*, *prime factorisation*, *square root*. Work through the example *Is 1728 a perfect cube?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Cube builders	Run the activity below. Move between groups and note who can explain the idea.	Build squares and cubes, group prime factor cards to test them, and trace a checking loop.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	An algorithm that tests whether n is a perfect square. Model the first card or step, then let pairs finish.	Trace it for n = 49 and n = 50. Where does k stop, and how many times does the loop run?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 8 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Code it on paper	Ask pairs to write the chapter's algorithm as a complete program, then trace it on three new inputs, including a tricky one.	Write and trace a full program; find an input that breaks a partner's program.
8	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Cube builders

1. Groups build squares (2 by 2, 3 by 3) and cubes (2 by 2 by 2) from unit cubes and record how many cubes each needs.
2. Give numbers as prime factor cards (for example 2, 2, 2, 3, 3, 3). Groups sort the cards into pairs (square test) and triples (cube test).
3. Ask: 'What is the smallest number of cards to add to make a cube?' Link to '56 x k is a cube'.
4. Pairs trace the is_square loop for 49 and 50 and count the loop runs.

Tip: Physical grouping of factor cards makes 'pairs for squares, triples for cubes' memorable.

Common mistakes to watch for

- **Confusing n^2 with $2n$**, or n^3 with $3n$.
- **Missing a prime factor** in the factorisation; check by multiplying back.
- **Off-by-one in the loop**: where k stops and how many times the loop runs.

Different paces

Support: Use numbers under 100 and factor trees drawn together.
Stretch: Find the smallest number that is both a perfect square and a perfect cube, other than 1 (64), and the next one (729).

Exit check and what to note

Two quick questions at the end of the chapter:

1. What is the smallest number you must multiply 180 by to get a perfect square? (Answer: 5 (180 = 2 x 2 x 3 x 3 x 5; 900 = 30 x 30))

2. Is 216 a perfect cube? (Answer: yes: 216 = 6 x 6 x 6)

Note in the journal: Beginning (needs help to test squares and cubes with prime factors); **Developing** (can test squares and cubes with prime factors with prompts or a partner); **Secure** (can test squares and cubes with prime factors and trace a checking algorithm alone); **Extending** (can find patterns in squares and cubes and explain why).

Answer key

1. 5: 64, 81, 100, 121 and 144
2. a) 2 (72 = 2 x 2 x 2 x 3 x 3; 144 = 12 x 12)
3. c) 49 (56 = 2 x 2 x 2 x 7, so two more 7s are needed)
4. d) 64 = 8 x 8 = 4 x 4 x 4
5. c) 7 (squares end in 0, 1, 4, 5, 6 or 9)
Build it: 49: k stops at 7 after 6 loops and it says 'square'. 50: k stops at 8 after 7 loops; 64 is not 50, so 'not a square'
Challenge: 469: $n^3 - (n - 1)^3 = 3 \times n \times (n - 1) + 1$, and 3 x 13 x 12 + 1 = 469 (check: 2197 - 1728)

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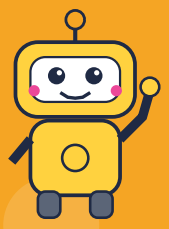
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TEACHER NOTES · CHAPTER 2 · LEVEL C6

Power Play

Coder skill: Exponents, doubling and counting combinations

Goes with NCERT maths, Class 8, chapter 2



At a glance	
Students will	use the laws of exponents, count combinations by multiplying choices and explain why doubling grows so fast.
Key words	**exponent** (how many times to multiply), **base** (the number being multiplied), **power** (a base with an exponent), **combination** (one way of choosing)
Materials	Sheets of paper to fold, outfit picture cards (shirts, trousers, shoes), a chessboard picture, calculators (optional)
Time	8 periods of 40 minutes
Link	Teach alongside chapter 2 of the NCERT Class 8 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *exponent*, *base*, *power*, *combination*. Work through the example *Folding a sheet of paper* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Fold, count and combine	Run the activity below. Move between groups and note who can explain the idea.	Record doubling as powers, list and count combinations, and run the doubling program.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	The chessboard story: 1 grain on the first square, then double on each square. Model the first card or step, then let pairs finish.	Trace it for the first 10 squares. What does it say? Can you see a shortcut?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 8 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Code it on paper	Ask pairs to write the chapter's algorithm as a complete program, then trace it on three new inputs, including a tricky one.	Write and trace a full program; find an input that breaks a partner's program.
8	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Fold, count and combine

1. Each student folds a sheet in half again and again, recording the layers after each fold. Write each as a power of 2.
2. Groups lay out 3 shirt, 2 trouser and 2 shoe cards and list every outfit systematically. Count them, then multiply 3 x 2 x 2.
3. Tell the chessboard story and run the doubling program for 10 squares.
4. Ask: 'Why does adding one character to a password matter so much?'

Tip: Ask students to predict before every count. Predictions about powers are usually far too small, which is the point.

Common mistakes to watch for

- **Multiplying base and exponent**: 2^5 is 32, not 10.
- **Adding choices instead of multiplying them** for combinations.
- **Adding exponents with different bases**: $2^3 \times 3^2$ is not 6^5.

Different paces

Support: Use only 2 choices of each item and draw a tree of outfits.
Stretch: How many 6-character passwords can be made from 26 small letters? Write it as a power (26^6).

Exit check and what to note

Two quick questions at the end of the chapter:

1. $3^4 \times 3^2 = 3^?$ (Answer: 6)

2. 3 shirts, 2 trousers and 2 pairs of shoes make how many outfits? (Answer: 12)

Note in the journal: Beginning (needs help to use the laws of exponents); **Developing** (can use the laws of exponents with prompts or a partner); **Secure** (can use the laws of exponents and count combinations by multiplying choices alone); **Extending** (can explain why doubling grows so fast and explain why).

Answer key

1. 8
2. a) 244: each term is a power of 3 plus 1 ($3^5 + 1$)
3. 12
4. c) 10,000 (10^4)
5. a) 2^{10} = 1,024, against 1,000
Build it: 1 + 2 + 4 + ... + 512 = 1,023, which is $2^{10} - 1$
Challenge: 7: there are 10 teams of 3 in all, and 3 of them contain both A and B

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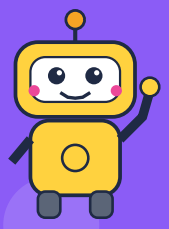
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TEACHER NOTES · CHAPTER 3 · LEVEL C6

A Story of Numbers

Coder skill: Number bases: binary, ternary and changing base

Goes with NCERT maths, Class 8, chapter 3



At a glance	
Students will	write numbers in base 2, 3 and 10, convert using repeated division and compare place value with Roman numerals.
Key words	**number base** (how many digits a system uses), **place value** (what a position is worth), **ternary** (base 3), **numeral** (a symbol for a number)
Materials	Paper strips cut to lengths in cm: binary set (1, 2, 4, 8, 16, 32, 64, one of each) and ternary set (1, 3, 9, 27, 81, two of each); base converter cards
Time	8 periods of 40 minutes
Link	Teach alongside chapter 3 of the NCERT Class 8 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *number base*, *place value*, *ternary*, *numeral*. Work through the example *50 inches with three kinds of pipe* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Measuring with pipes	Run the activity below. Move between groups and note who can explain the idea.	Measure lengths with decimal, binary and ternary strips and check with the base-conversion algorithm.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	One algorithm that converts to any base. Model the first card or step, then let pairs finish.	Trace it for n = 100 and b = 3. Write each n and each remainder.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 8 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Code it on paper	Ask pairs to write the chapter's algorithm as a complete program, then trace it on three new inputs, including a tricky one.	Write and trace a full program; find an input that breaks a partner's program.
8	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Measuring with pipes

1. As in CBSE's handbook activity, groups 'measure' heights with paper-strip pipes. Start with decimal strips (1 and 10, nine of each).
2. Measure the same lengths with binary strips (one of each power of 2) and write the binary number.
3. Repeat with ternary strips (two of each power of 3) and write the base-3 number.
4. Groups check each answer with the repeated-division algorithm.

Tip: Scale heights down (for example 66 cm for a 66-inch person) so the strips fit on a desk.

Common mistakes to watch for

- **Using a digit too big for the base**, such as 2 in binary or 3 in base 3.
- **Reading the remainders in the wrong order**; read from last to first.
- **Mixing place values**: base-3 places are 1, 3, 9, 27, not 1, 3, 6, 9.

Different paces

Support: Convert numbers under 32 to binary only.
Stretch: Convert 100 to base 5 (400) and check it with strips of lengths 1, 5 and 25.

Exit check and what to note

Two quick questions at the end of the chapter:

1. What is 13 in base 3? (Answer: 111 (9 + 3 + 1))

2. What is the binary number 101101 in decimal? (Answer: 45)

Note in the journal: Beginning (needs help to write numbers in base 2, 3 and 10); **Developing** (can write numbers in base 2, 3 and 10 with prompts or a partner); **Secure** (can write numbers in base 2, 3 and 10 and convert using repeated division alone); **Extending** (can compare place value with Roman numerals and explain why).

Answer key

1. a) 1010
2. b) 4
3. 34 (27 + 0 + 6 + 1)
4. a) MCMXLVII
5. c) 6: 32 + 16 + 8 + 4 + 2 + 1
Build it: 100: rem 1, n = 33; rem 0, n = 11; rem 2, n = 3; rem 0, n = 1; rem 1, n = 0. Read backwards: 10201
Challenge: A switch has two states, on and off, so two digits are easy to store reliably. 2222 in base 3 = 54 + 18 + 6 + 2 = 80

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TEACHER NOTES · CHAPTER 4 · LEVEL C6

4 Quadrilaterals

Coder skill: Classify shapes with if-else rules

Goes with NCERT maths, Class 8, chapter 4



At a glance	
Students will	sort quadrilaterals by their properties, use the angle sum of 360° and write a classifier with if-else rules.
Key words	**parallelogram** (two pairs of parallel sides), **rhombus** (four equal sides), **trapezium** (one pair of parallel sides), **diagonal** (joins opposite corners)
Materials	Shape cards (square, rectangle, rhombus, parallelogram, trapezium, kite), property cards, string loops for a family diagram
Time	8 periods of 40 minutes
Link	Teach alongside chapter 4 of the NCERT Class 8 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *parallelogram*, *rhombus*, *trapezium*, *diagonal*. Work through the example *Properties decide the name* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Property card sort	Run the activity below. Move between groups and note who can explain the idea.	Match properties to shapes, build a family diagram, and test an if-else shape classifier.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A quadrilateral classifier written with if-else. Model the first card or step, then let pairs finish.	Run it for: (a) 4 equal sides with angles 60° and 120°; (b) only one pair of parallel sides; (c) opposite sides equal and all angles 90°.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 8 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Code it on paper	Ask pairs to write the chapter's algorithm as a complete program, then trace it on three new inputs, including a tricky one.	Write and trace a full program; find an input that breaks a partner's program.
8	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Property card sort

1. Groups place property cards (2 pairs of parallel sides, all angles 90°, all sides equal, diagonals bisect each other) next to every shape they fit.
2. Build a 'family tree' on the floor with string loops: parallelograms inside quadrilaterals, rectangles and rhombuses inside parallelograms, squares in both.
3. A 'human classifier' asks the if-else questions from the book; others hold up shape cards to test it.
4. Ask: 'Which question splits the shapes best first?'

Tip: Ask 'Is a square a rectangle?' early. The debate shows that classes are defined by properties.

Common mistakes to watch for

- **Thinking a square is not a rectangle.** It meets every rectangle property.
- **Calling any shape with equal sides a square**; a rhombus has equal sides too.
- **Angle-sum slips**: a quadrilateral's angles add up to 360°.

Different paces

Support: Sort with two properties only: parallel sides and right angles.
Stretch: Add the kite to the classifier and find the fewest yes/no questions that identify all six shapes.

Exit check and what to note

Two quick questions at the end of the chapter:

1. A quadrilateral has all sides equal and diagonals that cross at 90°, but its angles are not 90°. What is it? (Answer: a rhombus)

2. Three angles of a quadrilateral are 90°, 90° and 120°. What is the fourth? (Answer: 60°)

Note in the journal: Beginning (needs help to sort quadrilaterals by their properties); **Developing** (can sort quadrilaterals by their properties with prompts or a partner); **Secure** (can sort quadrilaterals by their properties and use the angle sum of 360° alone); **Extending** (can write a classifier with if-else rules and explain why).

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Answer key

1. c) rhombus
2. 75°
3. a) every square is a rectangle
4. 115° (neighbouring angles add up to 180°)
5. 9: 4 small squares, 4 rectangles of 2 squares and 1 big square
Build it: (a) rhombus (b) trapezium (c) rectangle
Challenge: A square: diagonals that bisect each other make a parallelogram, equal diagonals make it a rectangle, and meeting at 90° makes it a rhombus

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TEACHER NOTES · CHAPTER 5 · LEVEL C7

5

Number Play

Coder skill: Divisibility rules and remainders as quick tests

Goes with NCERT maths, Class 8, chapter 5



At a glance	
Students will	use divisibility rules, work with remainders and trace a divisibility algorithm.
Key words	**divisible** (divides with no remainder), **remainder** (what is left over), **divisibility rule** (a shortcut test), **multiple** (in the times table)
Materials	Bulb cards labelled with divisors (3, 4, 8, 9, 11), number cards, calculators for checking
Time	8 periods of 40 minutes
Link	Teach alongside chapter 5 of the NCERT Class 8 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *divisible*, *remainder*, *divisibility rule*, *multiple*. Work through the example *Testing 7,392 without dividing* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Divisibility bulbs	Run the activity below. Move between groups and note who can explain the idea.	Apply divisibility rules as quick tests, check them by division, and use remainders in cycles.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	The divisibility test for 11 as an algorithm. Model the first card or step, then let pairs finish.	Trace it for 9,174 and for 5,213.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 8 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Code it on paper	Ask pairs to write the chapter's algorithm as a complete program, then trace it on three new inputs, including a tricky one.	Write and trace a full program; find an input that breaks a partner's program.
8	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Divisibility bulbs

1. Five students are 'bulbs', each with a divisor card. Show a number; a bulb 'lights' (stands) if its divisibility rule says yes.
2. The class checks each bulb's decision with the rule aloud, then with a calculator.
3. Groups work out which bulbs light for 7,392, 4,527 and 9,174.
4. Use remainders as a calendar: 'What day is 100 days from Monday?'

Tip: Bulb students must say their rule aloud. Saying the algorithm is how it becomes theirs.

Common mistakes to watch for

- **Applying the rule for 3 to 9**: a digit sum of 21 means divisible by 3, not by 9.
- **Alternating sum signs** for 11: start with + on the leftmost digit and keep alternating.
- **Remainder slips**: the remainder is always less than the divisor.

Different paces

Support: Use the rules for 2, 5 and 10, then add 3.
Stretch: Explain why the rule for 9 works, using 10 = 9 + 1 and 100 = 99 + 1.

Exit check and what to note

Two quick questions at the end of the chapter:

1. Is 8,712 divisible by 11? (Answer: yes: 8 - 7 + 1 - 2 = 0 (8,712 = 11 x 792))

2. What is the remainder when 50 is divided by 7? (Answer: 1)

Note in the journal: Beginning (needs help to use divisibility rules); **Developing** (can use divisibility rules with prompts or a partner); **Secure** (can use divisibility rules and work with remainders alone); **Extending** (can trace a divisibility algorithm and explain why).

Answer key

1. a) 4,527 (digit sum 18)
2. 2 (7 x 146 = 1,022)
3. b) 54
4. b) 12, the lowest common multiple
5. b) Wednesday (100 = 14 x 7 + 2)
Build it: 9,174: 9 - 1 + 7 - 4 = 11, divisible (9,174 = 11 x 834). 5,213: 5 - 2 + 1 - 3 = 1, not divisible
Challenge: 301: numbers 1 more than a multiple of 60 are 61, 121, 181, 241, 301, ...; the first divisible by 7 is 301 = 7 x 43

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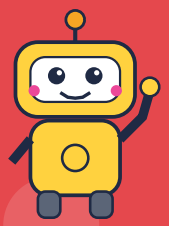
6

TEACHER NOTES · CHAPTER 6 · LEVEL C7

We Distribute Yet Things Multiply

Coder skill: The distributive property and identities for fast calculation

Goes with NCERT maths, Class 8, chapter 6



At a glance	
Students will	use the distributive property, use identities to calculate quickly and expand two brackets with an area model.
Key words	**distributive property** (a(b + c) = ab + ac), **identity** (true for every number), **expand** (multiply out the brackets), **factorise** (write as a product)
Materials	Squared paper, scissors, area-model templates, mental-maths cards
Time	8 periods of 40 minutes
Link	Teach alongside chapter 6 of the NCERT Class 8 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *distributive property*, *identity*, *expand*, *factorise*. Work through the example *23 x 47 with an area model* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Area model race	Run the activity below. Move between groups and note who can explain the idea.	Build area models, expand two brackets, and race with identities for fast calculation.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A function that multiplies two numbers using squares. Model the first card or step, then let pairs finish.	Use it for times(54, 46) and times(61, 59). Check one answer the long way.
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 8 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Code it on paper	Ask pairs to write the chapter's algorithm as a complete program, then trace it on three new inputs, including a tricky one.	Write and trace a full program; find an input that breaks a partner's program.
8	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Area model race

1. Groups draw and cut area models for 23 x 47 and 36 x 14, labelling each part.
2. Use the models to write the expansions, such as (20 + 3)(40 + 7).
3. Run a mental-maths race using identities: 99 x 101, 48^2, 103 x 97.
4. Groups explain each quick method with an identity on the board.

Tip: Ask a student who used the long method to check a quick answer. Agreement builds trust in identities.

Common mistakes to watch for

- **Missing a cross term**: $(a + b)^2$ is $a^2 + 2ab + b^2$, not $a^2 + b^2$.
- **Sign slips** in $(a - b)^2$.
- **Using an identity that does not match** the numbers, such as $a^2 - b^2$ for 47 x 52.

Different paces

Support: Expand a(b + c) with numbers before using letters.
Stretch: Use (a + b)(a - b) to find 1,005 x 995 quickly (999,975).

Exit check and what to note

Two quick questions at the end of the chapter:

1. Work out 99 x 101 quickly. (Answer: 9,999 ($100^2 - 1^2$))

2. Expand (x + 2)(x + 7). (Answer: $x^2 + 9x + 14$)

Note in the journal: Beginning (needs help to use the distributive property); **Developing** (can use the distributive property with prompts or a partner); **Secure** (can use the distributive property and use identities to calculate quickly alone); **Extending** (can expand two brackets with an area model and explain why).

Answer key

1. 9,996 (10,000 - 4)
2. a) 6x + 15
3. 29 (49 - 2 x 10)
4. a) 360 = (47 + 43)(47 - 43) = 90 x 4
5. a) $x^2 + 8x + 15$
Build it: 54 x 46: $50^2 - 4^2$ = 2,500 - 16 = 2,484. 61 x 59: $60^2 - 1^2$ = 3,599
Challenge: Otherwise (a + b) ÷ 2 is not a whole number. It still works with halves: $5.5^2 - 1.5^2$ = 30.25 - 2.25 = 28

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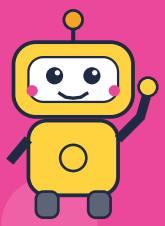
7

TEACHER NOTES · CHAPTER 7 · LEVEL C7

Proportional Reasoning

Coder skill: Ratios, the unitary method and scaling

Goes with NCERT maths, Class 8, chapter 7



At a glance	
Students will	simplify and compare ratios, use the unitary method and tell direct from inverse proportion.
Key words	**ratio** (a comparison like 3 : 5), **proportion** (two equal ratios), **unitary method** (find the value for one first), **inverse proportion** (one up, the other down)
Materials	Station cards: recipes, maps with scales, painters and days, price lists; calculators (optional)
Time	8 periods of 40 minutes
Link	Teach alongside chapter 7 of the NCERT Class 8 maths textbook.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *ratio*, *proportion*, *unitary method*, *inverse proportion*. Work through the example *Direct and inverse proportion* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Proportion stations	Run the activity below. Move between groups and note who can explain the idea.	Solve proportion problems at stations, decide direct or inverse, and write a scaling function.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A function that scales a recipe. Model the first card or step, then let pairs finish.	What does the program say? Is a recipe an example of direct or inverse proportion?
5	Practise and extend	Go over the questions students found hardest, then run the stretch task with the whole class.	Correct their answers, explain one fix to a partner, then try the stretch task.
6	Puzzle period	Set two or three puzzles from the matching chapter of CBSE's Class 8 handbook. Pairs solve them, then present one method.	Solve handbook puzzles in pairs and explain one method to the class.
7	Code it on paper	Ask pairs to write the chapter's algorithm as a complete program, then trace it on three new inputs, including a tricky one.	Write and trace a full program; find an input that breaks a partner's program.
8	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Proportion stations

1. Set up four stations: scale a recipe, use a map scale, share work among painters, compare prices.
2. Groups rotate, solving each problem with the unitary method and writing the 'value for one' first.
3. At each station, groups label the problem 'direct' or 'inverse' proportion.
4. Groups write a scale(amount, from, to) function and test it on their recipe.

Tip: Ask 'If one quantity doubles, what happens to the other?' It decides direct or inverse every time.

Common mistakes to watch for

- **Treating inverse proportion as direct**: more painters means fewer days.
- **Ratios in the wrong order**: 3 : 5 is not 5 : 3.
- **Mixing units** before comparing ratios.

Different paces

Support: Use whole-number answers and the unitary method step by step.
Stretch: A car uses 6 litres for 90 km. How far on 15 litres? (225 km) Write it as a function.

Exit check and what to note

Two quick questions at the end of the chapter:

1. 5 pens cost Rs 60. How much do 8 pens cost? (Answer: Rs 96)

2. 12 workers take 10 days for a job. How long would 15 workers take? (Answer: 8 days)

Note in the journal: Beginning (needs help to simplify and compare ratios); **Developing** (can simplify and compare ratios with prompts or a partner); **Secure** (can simplify and compare ratios and use the unitary method alone); **Extending** (can tell direct from inverse proportion and explain why).

Answer key

1. a) 3 : 4
2. 37.5 km
3. 20 (10, 15 and 20)
4. b) 16 days (96 worker-days ÷ 6)
5. a) 12 : 20
Build it: 750 g of rice and 5 cups of milk; direct proportion
Challenge: 10 litres: there are 25 litres of milk and 15 of water, so 10 more litres of water are needed

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AI1

TEACHER NOTES · AI CHAPTER 1 · LEVEL C8

AI Project Lifecycle

AI skill: Define, collect, train, test and deploy

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 1



At a glance	
Students will	name the stages of an AI project, clean a small dataset and measure accuracy and decide whether to deploy.
Key words	**problem statement** (what the AI must do), **data cleaning** (fixing messy data), **model** (what the AI learns from data), **deployment** (putting it to real use)
Materials	Dataset cards: 24 past deliveries (distance, weather, traffic, late or on time), including 2 duplicates and 2 with missing values; scoring slips
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 1.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *problem statement*, *data cleaning*, *model*, *deployment*. Work through the example *Cleaning a small dataset before training* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: An AI project on paper	Run the activity below. Move between groups and note who can explain the idea.	Run every stage of the AI project lifecycle by hand: define, clean, train a rule, test and decide.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	An online shop wants an AI that predicts whether a delivery will be late. Plan the project. Model the first card or step, then let pairs finish.	The model gets 34 of 40 test orders right. What is its accuracy? The shop needs 90%. Should it deploy the model?
5	Data cleaning clinic	Give a messy table with missing, duplicate, impossible and irrelevant values. Groups clean it and explain each fix.	Clean a dataset and justify every change.
6	Deploy or not?	Ask: 'Your rule is 60% accurate. Would you deploy it? Who is affected by the mistakes?' Hold a short debate.	Decide on deployment with reasons, and suggest how to improve the model.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: An AI project on paper

1. Read the problem: predict whether a delivery will be late. Groups predict which features will matter.
2. Groups clean the deck: remove duplicates and cards with missing values. Then put 5 cards aside, unseen, as test cards.
3. Using the other cards, groups find a rule, such as 'heavy traffic and over 10 km means late', and test it on the 5 unseen cards.
4. Order jumbled stage cards (define, collect, prepare, train, test, deploy) and match each stage to what they did.

Tip: Shuffle the duplicates and missing-value cards in, so cleaning is a real step, not a pretend one.

Common mistakes to watch for

- **Testing on the training cards.** A rule always looks good on the examples it was made from.
- **Skipping cleaning.** Duplicates make one order count twice.
- **Deploying because it is 'mostly right'.** Decide the needed accuracy before testing.

Different paces

Support: Give groups a starting rule and ask them only to clean, test and count.
Stretch: Compare two rules on the same test cards, and work out accuracy for 'late' and 'on time' cards separately.

If you have computers (optional)

Run the full lifecycle with a no-code tool: define two classes, collect and label images, train, test on new images and decide whether it is good enough.

Exit check and what to note

Two quick questions at the end of the chapter:

1. A rule gets 4 of 5 test cards right. What is its accuracy? (Answer: 80%)

2. Why must the test cards be kept apart from the training cards? (Answer: to check the rule on data it has not seen)

Note in the journal: Beginning (needs help to name the stages of an AI project); **Developing** (can name the stages of an AI project with prompts or a partner); **Secure** (can name the stages of an AI project and clean a small dataset alone); **Extending** (can measure accuracy and decide whether to deploy and explain why).

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Answer key

1. a) define the problem
2. 75%
3. a) the car wash: it is automation
4. a) to check the model on data it has not seen
5. a) remove duplicates
Build it: 85%. Not yet: it should improve the data (for example more orders in bad weather) and retrain
Challenge: Open answer, for example: define 'real or fake headline'; collect checked real and fake headlines; clean and label them; train; test on new headlines; refine and only deploy with human checks

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AI2

TEACHER NOTES · AI CHAPTER 2 · LEVEL C8

AI and Its Applications

AI skill: AI for people and planet; training no-code models

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 2



At a glance	
Students will	give examples of AI helping society, describe how a no-code model is trained and improve a model by improving its data.
Key words	**no-code tool** (builds models without coding), **class** (one label a model can choose), **sensor** (a device that measures), **early warning** (an alert before harm)
Materials	Waste picture cards (plastic, paper, organic) photographed against one background and against varied backgrounds, case-study sheets
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 2.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *no-code tool*, *class*, *sensor*, *early warning*. Work through the example *Training a waste-sorting model with a no-code tool* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: The waste lab, on cards	Run the activity below. Move between groups and note who can explain the idea.	Act out training and testing a waste classifier, and see how varied data improves it.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	Plan a no-code voice-command model. Model the first card or step, then let pairs finish.	It was tested 30 times: 'lights on' 9 of 10 right, 'lights off' 6 of 10, noise 10 of 10. Which class needs more data? What is the overall accuracy?
5	AI for India gallery	Groups read short cases (chest X-ray screening for TB, crop advice from photos, air-quality forecasts, wildlife camera traps) and present the data each uses.	Present one real application: the problem, the data and one risk.
6	Voice lab plan	Plan a voice-command model: classes, number of samples, variety of voices, and how to test it.	Write a data plan for a voice model and predict where it will fail.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: The waste lab, on cards

1. This is the card version of CBSE's no-code waste-classification lab. One student is the 'model' and may only learn from the cards shown.
2. Train the 'model' with cards that all share one background. Test with cards on new backgrounds: note the mistakes.
3. Retrain with varied cards and test again. Compare the scores.
4. Ask: 'What did the model really learn the first time?' (sometimes the background, not the object)

Tip: Choose a student who is happy to be 'wrong' on purpose. The class learns most from the model's mistakes.

Common mistakes to watch for

- ✓ **Training on very similar examples**, so the model learns the background instead of the object.
- ✓ **Testing with the training cards** and believing the high score.
- ✓ **Thinking no-code means no thinking.** Choosing and checking data is the hard part.

Different paces

Support: Use two classes only (plastic and paper).
Stretch: Design a test set that would catch a model that learnt backgrounds instead of objects.

If you have computers (optional)

Run CBSE's waste-classification and voice-command labs in a no-code tool, then test with new items, new backgrounds and different voices.

✓ Exit check and what to note

Two quick questions at the end of the chapter:

1. A waste model works in the classroom but not in the canteen. What is the best fix? (Answer: add training photos with varied backgrounds and light)

2. What does a no-code AI tool let you do? (Answer: train a model without writing code)

Note in the journal: Beginning (needs help to give examples of AI helping society); **Developing** (can give examples of AI helping society with prompts or a partner); **Secure** (can give examples of AI helping society and describe how a no-code model is trained alone); **Extending** (can improve a model by improving its data and explain why).

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Answer key

1. a) the environment
2. a) healthcare
3. a) train a model without writing code
4. a) its training photos all had the same background
5. a) train it with samples from many different voices
Build it: 'lights off' needs more varied samples; overall 25 of 30, about 83%
Challenge: Open answer, for example: a camera model that spots overflowing bins, trained on labelled photos of full and empty bins

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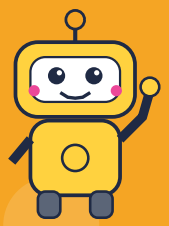
AI3

TEACHER NOTES · AI CHAPTER 3 · LEVEL C8

Data and Fairness in AI

AI skill: Spot unbalanced data and rebalance it

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 3



At a glance	
Students will	spot over- and under-represented data, explain how unbalanced data leads to unfair results and plan how to rebalance a dataset.
Key words	**representative** (includes every group fairly), **over-represented** (too many examples), **under-represented** (too few examples), **rebalance** (even out the data)
Materials	Training cards: 18 cricket bats and 2 badminton rackets; a balanced test set; survey data cards (sport by group)
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 3.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *representative*, *over-represented*, *under-represented*, *rebalance*. Work through the example *A sports-recommendation model (practice data)* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Bat or racket?	Run the activity below. Move between groups and note who can explain the idea.	Show how unbalanced data skews predictions, measure each class, rebalance and test again.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A rebalancing plan for an image dataset. Model the first card or step, then let pairs finish.	Classes: dogs 300, cats 100, rabbits 40. How many more cats and rabbits are needed so that no class is less than half the biggest?
5	Translation and stereotypes	Show gender-neutral sentences in Hindi or another language and how a translator might add 'he' or 'she'. Discuss why.	Explain how stereotypes in training text appear in AI output.
6	Rebalancing plan	Give class counts (dogs 300, cats 100, rabbits 40). Groups plan how many examples to add and what variety they need.	Write a rebalancing plan with numbers and reasons.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Bat or racket?

1. As in CBSE's handbook example, a 'model' student learns only from the lopsided training cards.
2. Test with a balanced set, including unclear pictures. Count right answers for bats and rackets separately.
3. Rebalance with more varied racket cards and test again.
4. Discuss the sports-recommendation data: should a model suggest a sport based on gender, or on a student's interests?

Tip: Keep the conversation on data, not on blaming people. The aim is to see how patterns in data become unfair outcomes.

Common mistakes to watch for

- **Thinking balanced numbers are enough.** Variety within each class matters too.
- **Using sensitive features** such as gender to make choices they should not decide.
- **Checking only overall accuracy**, which hides a weak class.

Different paces

Support: Work with two classes and counts under 50.
Stretch: Describe a dataset that is balanced in numbers but still unfair, and how to fix it.

If you have computers (optional)

Run CBSE's cat-and-dog training activity with a no-code tool: test, re-curate the training set, retrain and compare.

Exit check and what to note

Two quick questions at the end of the chapter:

1. A dataset has 200 dog photos and 50 cat photos. How many more cat photos make cats at least half as many as dogs? (Answer: 50 more)

2. What does 'representative data' mean? (Answer: data that includes every group and situation fairly)

Note in the journal: Beginning (needs help to spot over- and under-represented data); **Developing** (can spot over- and under-represented data with prompts or a partner); **Secure** (can spot over- and under-represented data and explain how unbalanced data leads to unfair results alone); **Extending** (can plan how to rebalance a dataset and explain why).

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Answer key

1. a) cricket bat, because that class dominates its training data
2. a) add more varied racket photos
3. a) its training data had more male doctors
4. a) not representative
5. 25%
Build it: Half of 300 is 150, so 50 more cats and 110 more rabbits
Challenge: Yes: 100 cat and 100 dog photos, but every cat is indoors and every dog outdoors, so the model learns the background; or all photos come from one region

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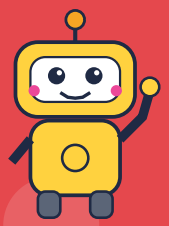
AI4

TEACHER NOTES · AI CHAPTER 4 · LEVEL C8

Ethics and Responsible AI

AI skill: Privacy, checking facts and human accountability

Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 4



At a glance	
Students will	protect my personal data, check information before sharing and explain why humans stay accountable for AI.
Key words	**privacy** (control over your personal data), **consent** (agreeing to share data), **misinformation** (false information that spreads), **accountability** (people answer for AI decisions)
Materials	App permission cards, claim cards for fact-checking, scenario cards for accountability, chart paper
Time	7 periods of 40 minutes
Link	Goes with the AI part of CBSE's Class 8 CT&AI handbook, chapter 4.

Period by period

Period	Focus	Teacher does	Students do
1	Learn and see	Read Robo's tip aloud and teach the words *privacy*, *consent*, *misinformation*, *accountability*. Work through the example *Which permissions should an app get?* on the board, asking for a prediction before each step.	Predict each step, then explain the example to a partner in their own words.
2	Hands-on: Permission audit and fact-check	Run the activity below. Move between groups and note who can explain the idea.	Audit app permissions, plan how to fact-check claims, and decide who is accountable for AI decisions.
3	Try it!	Set questions 1-5. After about 15 minutes, pairs swap books and check with the answer key. Discuss the hardest question together.	Answer questions 1-5. Explain one answer to the class.
4	Now you build it	A responsible-AI checklist written as rules. Model the first card or step, then let pairs finish.	Use the checklist: an AI homework helper asks for your location and contacts, and gives you a 'fact' for your project.
5	Fact-check race	Groups get three claims and a list of trusted sources. They decide true, false or unsure, and show how they checked.	Check claims against trusted sources and explain their method.
6	Who is accountable?	Debate: 'If an AI makes a harmful mistake, who is responsible?' Each side gives reasons.	Argue a position with reasons and agree on human accountability.
7	Challenge and check	Give the challenge to early finishers, then solve it together. Run the exit check and note levels in the journal.	Try the challenge, colour 'How did I do?' and answer the exit check.

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Hands-on activity: Permission audit and fact-check

1. Groups sort app permission cards into 'allow', 'allow while using' and 'deny', giving a reason for each.
2. Give claim cards (a shocking headline, a doctored photo description, a real fact). Groups decide how they would check each one.
3. Act out a hospital scenario: an AI flags a patient as high risk. Who decides, and who is responsible if it is wrong?
4. Groups write a five-rule 'responsible AI' charter for the class.

Tip: Use made-up apps and claims. The habit of asking 'does it need this?' and 'how do I know?' is what matters.

Common mistakes to watch for

- **Allowing every permission** to make an app work faster.
- **Sharing first and checking later.**
- **Blaming 'the computer'** instead of the people who made or used the system.

Different paces

Support: Use sentence frames: 'This app needs ___ because ___.'
Stretch: Read a real app's permission list (with an adult) and decide which permissions it really needs.

If you have computers (optional)

Open a phone or browser's privacy settings together on the projector and review which apps can use location and camera.

Exit check and what to note

Two quick questions at the end of the chapter:

1. A torch app asks for your location. Should you allow it? (Answer: no: a torch does not need your location)

2. Who is responsible for decisions made with AI? (Answer: people: the makers and the people or organisation using it)

Note in the journal: Beginning (needs help to protect my personal data); **Developing** (can protect my personal data with prompts or a partner); **Secure** (can protect my personal data and check information before sharing alone); **Extending** (can explain why humans stay accountable for AI and explain why).

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Answer key

1. a) deny it: a torch does not need contacts
2. a) check trusted news sources
3. a) the doctor, after reviewing the AI's suggestion
4. a) your home address
5. a) no: it must be fixed for that group
Build it: Deny location and contacts; check the fact with a trusted source before using it, and say that you used AI
Challenge: People: the makers and the organisation using it are accountable, because a machine cannot be held responsible

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P1

TEACHER GUIDE · PROJECT 1 · 20 HOURS · LEVEL C8

Sports for Everyone

Project skill: Survey, analyse with ratios and recommend changes

CBSE: 40 project hours a year in Class 8, as two projects of 20 hours



At a glance

Big question	Which students in our school play sport regularly, and what change could get more students active?
Students will	design an anonymous survey, collect and clean data, analyse it with percentages and ratios, chart it honestly and recommend changes backed by evidence
Groups	Teams of 3 or 4, with roles that rotate at the halfway point
Materials	Survey forms, tally sheets, data sheet pads (kit), squared paper, chart paper, calculators (optional)
Time	30 periods of 40 minutes (about 2 periods a week for 10 weeks)
Evidence	The survey and consent note, cleaned data, percentage and ratio calculations, two charts, a recommendation with a test plan, and the completed rubric

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Week by week

Weeks	Stage	Students do	Teacher does
1	Define	Agree the question and what 'plays regularly' means.	Form teams and roles. Agree the question and what 'plays regularly' means (for example, at least three times a week).
2	Design	Write 5-6 clear survey questions. Plan how to ask permission and keep answers anonymous.	Check survey questions for clarity and bias. Approve a short consent note explaining how answers will be used.
3	Collect	Survey at least 60 students from different classes.	Arrange time with class teachers for the survey. Check that no names are collected.
4	Clean	Remove duplicates and blank forms. Note anything odd.	Show how to remove duplicates and blank forms, and to note odd answers instead of deleting them silently.
5	Analyse	Work out percentages and ratios for each class, and the most common reasons.	Revise percentages and ratios. Check one calculation per team.
6	Show	Draw a bar chart for each class and a pie chart of reasons, with honest scales.	Check chart choices and that scales start at 0.
7	Find patterns	Write three observations and one conclusion, with its limits.	Ask for three observations and one conclusion, each with its limit.
8	Recommend	Suggest one or two changes and how you would test them.	Check that every recommendation points to the data, and that teams plan how to test it.
9	Present	Present to your class or sports teacher in 3 minutes.	Invite the sports teacher to the presentations if possible.
10	Reflect	Score your work with the rubric and write what you would change.	Students score themselves with the rubric; you score too. Discuss any differences.

Setting up

- ✓ Ask permission from class teachers before surveying their students.
- ✓ Keep the survey anonymous: class and answers only, no names.
- ✓ Do not recommend or judge sports by gender or any personal characteristic.
- ✓ Agree the survey questions before collecting, and do not change them halfway.

Team roles

Give each student a role and rotate roles at the halfway point, so everyone collects, checks, builds and presents.

Survey lead: designs the questions and runs the survey.

Data cleaner: checks forms for duplicates, blanks and odd answers.

Analyst: works out percentages and ratios and draws the charts.

Presenter: leads the recommendation and the talk.

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Common pitfalls

- **Comparing counts from groups of different sizes** instead of percentages.
- **Leading questions**, such as 'Don't you think sport is boring?'.
- **Recommendations not backed by the data**.

Different paces

Support: Give a ready-made survey and partly filled tally sheet.
Stretch: Run the recommendation for a month (with permission) and survey again to see if participation changed.

If you have computers (optional)

Collect answers with a free online form tool and analyse them in a spreadsheet with percentage formulas and charts.

Marking with the rubric

Score each row from 1 to 4 using the survey, cleaned data, calculations, charts and talk; the total is out of 24. Give credit for honest limits and for a clear plan to test the recommendation.

What we check	What excellent (4) looks like
Question	A clear question about real participation.
Survey	Fair, anonymous questions; permission asked; 60 or more answers.
Analysis	Percentages and ratios correct; patterns found.
Charts	The right chart types, honest scales and clear labels.
Recommendation	Changes backed by the data, with a plan to test them.
Teamwork and talk	Work shared fairly; a clear presentation.

What a strong project sounds like

'72% of Class 6 play sport regularly, but only about 36% of Class 8. The most common reason in Class 8 was no time after school. We recommend a 20-minute lunchtime games slot for Class 8, and we will survey again after a month to see if it helps.'

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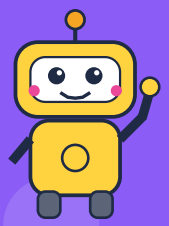
P2

TEACHER GUIDE · PROJECT 2 · 20 HOURS · LEVEL C8

Fair AI Audit

Project skill: Test a classifier for bias, rebalance and report

CBSE: 40 project hours a year in Class 8, as two projects of 20 hours



At a glance	
Big question	Is a picture classifier equally good for everything it is used on, and can we make it fairer?
Students will	test a picture classifier group by group, investigate its training data, rebalance it, retest on the same test set and write an audit report
Groups	Teams of 3 or 4, with roles that rotate at the halfway point
Materials	AI audit picture cards (kit) or photos for a no-code tool, data sheet pads, chart paper
Time	30 periods of 40 minutes (about 2 periods a week for 10 weeks)
Evidence	The fixed test set, the audit record, accuracy for each group before and after, the training-data count, a chart and the written audit report

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Week by week

Weeks	Stage	Students do	Teacher does
1	Define	Choose the classifier and the groups to compare (for example cats and dogs, indoor and outdoor).	Form teams and roles. Help each team choose a classifier and the groups to compare (cats and dogs, indoor and outdoor).
2	Prepare	Build a balanced test set: 20 unseen examples per group.	Check that each test set is balanced and unseen. Collect a copy so it cannot change later.
3	Test	Run every test example and record the prediction and the true label.	Teams run every test example and record predictions. Check the audit record format.
4	Measure	Work out the accuracy for each group and the gap between them.	Check accuracy for each group and the gap. Ask: 'What does the overall score hide?'
5	Investigate	Count the training examples per group. How are they alike, and how are they different?	Help teams count and describe the training data: what is missing or rare?
6	Rebalance	Add varied examples of the weak group: other backgrounds, lights and types.	Check that new examples add variety (backgrounds, light, types), not just more of the same.
7	Retest	Retrain and test again on the same test set.	Make sure teams retest on exactly the same test set as before.
8	Compare	Make a before-and-after table and a bar chart.	Help teams build a before-and-after table and a fair chart.
9	Report	Write the audit report: findings, fix, remaining limits and advice.	Teams write the audit report: findings, fix, remaining limits and advice. Read drafts and give feedback.
10	Reflect	Score your work with the rubric. Where could such gaps harm people?	Students score themselves with the rubric; you score too. Discuss where gaps like this could harm people.

Setting up

- ✓ Use pictures of objects or animals only: no photos of people.
- ✓ Use only images the team is allowed to use.
- ✓ Fix the test set before any changes, and keep it the same for the retest.
- ✓ Agree what counts as a 'group' for the audit before testing.

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★ Team roles

Give each student a role and rotate roles at the halfway point, so everyone collects, checks, builds and presents.

Test manager: keeps the test set fixed and runs the tests.

Data auditor: counts and describes the training data.

Rebalancer: finds and adds varied new examples.

Report writer: writes the audit report and leads the talk.

Common pitfalls

- ✓ **Changing the test set** between the first test and the retest.
- ✓ **Reporting only overall accuracy** and hiding the gap between groups.
- ✓ **Claiming the model is 'fixed'** when a gap remains.

Different paces

Support: Give a ready-made card classifier and test set; teams measure, investigate and rebalance.
Stretch: Audit a second kind of gap (indoor against outdoor photos) and report both.

If you have computers (optional)

Train the classifier in a no-code tool, export or record its predictions on the test set, and compare before and after rebalancing.

Marking with the rubric

Score each row from 1 to 4 using the audit record, data counts, retest and report; the total is out of 24. Give full credit to teams that report remaining gaps honestly.

What we check	What excellent (4) looks like
Audit question	Clear groups to compare and a fair, balanced test set.
Testing	Accuracy measured for each group on unseen data.
Investigation	Training data counted and its gaps explained.
Improvement	Data rebalanced and the model retested on the same test set.
Report	Findings, fix and remaining limits stated honestly.
Teamwork and talk	Work shared fairly; a clear presentation.

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What a strong project sounds like

'Before, our classifier was right for 95% of dogs but only 55% of cats: a 40-point gap hidden in 75% overall. Most cat photos were indoors. After adding 90 varied cat photos, cats rose to 85% and the gap fell to 5 points. We still have no photos taken at night, so we cannot say it works then.'

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Observation journal

Class 8 ____ Section ____ Term ____ Chapters _____. Circle one level per skill: **B**eginning, **D**eveloping, **S**ecure, **E**xtending. You may photocopy this page.

Student	Algorithms	Numbers	AI projects	Projects	Notes
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	
	B D S E	B D S E	B D S E	B D S E	

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Term summary for the report card

One copy per child each term. Fill it from the observation journal and the child's self-checks. You may photocopy this page.

Name _____ Class _____ Section _____ Term _____

Area	Level (circle one)	Teacher comment
Computational thinking (overall)	B D S E	
Algorithms and classifiers	B D S E	
Number structure (powers, bases, identities)	B D S E	
AI projects, fairness and ethics	B D S E	
Projects (rubric score)	B D S E	

Writing comments

Write one strength and one next step, in words a parent understands. For example: *Explains steps clearly and fixes bugs in routes; next step: spotting the core of a pattern.*

Class teacher _____ Date _____

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